

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Moodify: A Personalized Music Recommendation Engine Built on Graphs, Hashing and Dynamic Data Structures

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

box 1 Despite the vast number of songs available on various music streaming platforms, finding the right At times it can be difficult to produce a song based on the listeners' preferences and moods. Search methods require users to search for a particular song, artist, or genre, and may not always find music in Their feeling may change as well, and the user might get fed up with the same old songs and look for other songs. Just like the ones they prefer. There is a need for a recommendation system which is able to organize and relate The reason for creating Moodify was to develop a personal recommendation system. Suggests songs based on the presently selected mood and also includes songs or artists related to it. In- stead of making use of complex machine-learning methods in the recommendation system, Moodify Makes use of a system based on rules, which is relatively easy to implement in C++. One of the main reasons for creating Moodify was to apply the concepts learned in Data Structures in a real-world The issue is that music recommendations make use of various data structures, including graphs used for representing relationships between songs/artists, hashing for faster access and storage of specific song details, and Dynamic data structures are used for the management and organisation of playlists. With Moodify it will be possible to demonstrate how data organization and algorithmic expertise can be combined to come up While improving one's understanding of C++ and data structures, efficient solutions can also be achieved.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

box 2 Music streaming services like Spotify or Apple Music recommend songs based on listening history, People like them, skip them, select artists, create playlists, and choose genres. However, the recommendations are generally repetitive or based on popularity, promotions, or other factors that have nothing to do with the current mood or immediate Preferences; although the commercial services make use of machine learning, big data, and algorithms, These techniques are not offered as part of the course. Moodify is a project intended to demonstrate a solution for the It looks at a smaller scale and establishes rule-based associations between songs. It is written in C++. The rule-based approach, graphs, hashing, and dynamic data structures are all used. Association The program creates connections between the songs with regard to their rules, moods, genres, artists, or other attributes. Going through a graph of audio items to get recommendations that meet the criteria. Graphs, Hash tables and dynamic data structures are used to implement various parts of the solution. The graph structure- The system uses the technique of repeatedly going through the audio items in order to obtain recommendations, and hash tables are employed for this purpose. storing and finding data, and dynamic data structures allow the program to modify the data during The execution [part], overall, shows that graphs, hashing, and dynamic data structures can be used together to implement a simple recommendation system for music

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The first step is to create a music recommender system called Moodify, where the software would use C++ to determine the mood and listening habits of a particular user and suggest the songs matching their preference. In this process, the suggested songs would be different for each and every user and won't be the same. This system would take into account the preferred artists, music listening history, liked songs, skipped songs, recently played songs, and various other parameters to recommend the songs as per the preferences of the user. Firstly, the system would create a song, artist, mood, and various other music related databases and then design the user profile with the features such as mood selection, searching, playlist, play, like, and skip. Another aim of the project is to show the capabilities of the C++ programming language and Data Structures by applying them to the development of Moodify as a practical application. Hashing will be used for storing and accessing the data related to songs and artists, linked list for organizing playlists, stack for storing the songs which were listened last. Moreover, queue will be applied to storing the songs to play next, priority queue to storing songs that will be recommended to users, and graphs for storing the relations between similar songs. After developing the application, we are going to put everything together using the recommendation engine, which will evaluate and order songs depending on the currently played mood and user preferences. Finally, we will test the application on different preferences and interactions in order to verify that the program can provide appropriate recommendations. In general, with this project, we are going to develop the music player with advanced functions and discover how C++ and Data Structures can be used together in practice.

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

1. Our "Moodify" project will be developed using C++, where we will maintain listening history, likes and skips of users by using file handling, and we will also develop a database which will consist of names of songs, artists and moods along with related songs. The database will help us arrange the available songs as per the mood of the user. 2) We will use different data structures for example for quick song and mood searches we will use hashing, for playlist we will use linked list, for upcoming songs queue will be used and priority queue for ranking recommendations 3) The system will determine the suitable songs for the user through an analysis of the user's mood, music listening style, and feedback. Scores will be assigned to each song, and playlists containing the best songs according to their scores will be generated. In addition to this, we will test the system with various users.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).

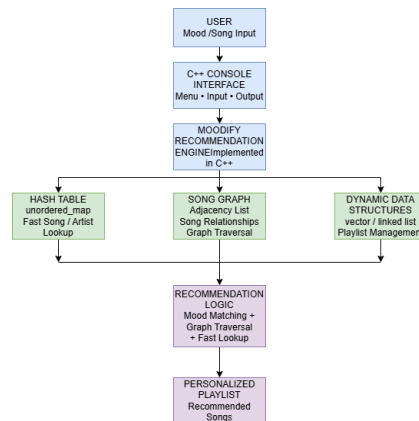


Figure 1: Overview of Moodify Music Recommendation System

Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

Our project's main deliverable will be a working music recommendation system built with C++ that enables users to obtain personalized song recommendations based on their current mood and listening habits. One of the key elements of the project will be to demonstrate how various **Data Structures** can be used in real-world application. Related songs will be displayed in graphs, songs and artists will be hashed for quick access, playlists will be linked lists, recently played songs will be in stacks, upcoming songs will be in queues and priority queues will be used to rank recommendations based on their scores. From the research paper we studied, we identified that the traditional popularity-based recommenders tend to recommend songs and artists that are popular in the overall population. This means that popular artists take over the recommendations and less popular or niche artists are ignored. To address this issue, we will adapt our recommender system to be more relevant to the user's personal preferences such as their listening history, likes and interests rather than just overall popularity. Overall, the project will be a practical demonstration of how C++ and Data Structures can be combined to build an interactive, efficient and personalized music recommendation system.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

The song dataset and the artist and mood associated with a song is defined. 2) Assume Moodify only suggests songs and does not actually stream or play or let you download them. 3) It is assumed that Moodify employs data-structure based recommendation system, and not any AI or machine learning model. 4) System also assumes that the input provided by the user (song/mood/artist) is valid. 5) It is assumed that the system utilizes data structures such as graphs, hashing, and dynamic data structures for storing, organizing, and retrieving music-related information. 6) the system is assumed to work with a fixed number of moods, instead of trying to work with every emotional state that is present. 6) Each song is assumed to have some mood associated with it. 7) It is assumed that each song has a unique id related to it.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

- 1) Article - An Analysis of Global and Regional Mainstreamness for Personalized Music Recommender Systems Published by - Markus Schedl and Christine Bauer Received 30 January 2018; Accepted 01 February 2018; Publication 20 April 2018 page 5,19.
link-<https://ieeexplore.ieee.org/document/11080174>
- 2) MoodPlay: Interactive Music Recommendation Based on Artists' Mood Similarity Authors: Ivana Andjelkovic, Denis Parra, John O'Donovan Published: 2018 Journal: International Journal of Human-Computer Studies Volume: 121 Pages: 142-159
link-<https://www.sciencedirect.com/science/article/abs/pii/S1071581918301654>
- 3) An Analysis of Global and Regional Mainstreamness for Personalized Music Recommender Systems Authors: Markus Schedl and Christine Bauer Journal: Journal of Mobile Multimedia Volume/Issue: Vol. 14, No. 1 Year: 2018 Pages: 95-112 according to the journal indexing record. Utrecht University
link-<https://ieeexplore.ieee.org/document/10495033>
- 4) Smoodi: Mood-Based Music Recommendation Player Authors: Seung Jae Lee, Jung Hyun Kim, Sung Min Kim, Won Young Yoo Published: July 2011 Conference: IEEE International Conference on Multimedia and Expo (ICME 2011) Pages: 1-4
link-<https://ieeexplore.ieee.org/document/6012116>
- 5) Emotion-Aware Music Recommendation Authors: Hieu Tran, Tuan Le, Anh Do, Tram Vu, Steven Bogaerts, Brian T. Howard Published: 2023 Conference: AAAI-23 Proceedings: Proceedings of the AAAI Conference on Artificial Intelligence Volume: 37, Issue 13 Pages: 16087-16095
link-<https://ojs.aaai.org/index.php/AAAI/article/view/26911>